

S14 – Edge Benchmark Results – Raspberry Pi 4

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Purpose

The purpose of this supplementary analysis is to report the measured inference performance of the calibrated HIMS-CDI model on a representative clinical-grade edge device, as referenced in Section 5.3 of the main manuscript. In addition, this section consolidates the computational resource estimates for all evaluated models presented previously in Table 4 of the main manuscript to provide a unified deployment-oriented comparison.

Hardware and Software Configuration

Hardware Platform

- **Device:** Raspberry Pi 4 Model B
- **Memory:** 4 GB RAM
- **Processor:** Quad-core ARM Cortex-A72 @ 1.5 GHz

Software Environment

- **Operating System:** Raspberry Pi OS Lite (64-bit, Debian Bullseye)
- **Runtime Environment:** ONNX Runtime 1.15 with Python 3.11
- **Model Export Sources:**
 - HIMS-CDI: Exported from scikit-learn
 - Baseline Models: Exported from PyTorch

Model Optimisation

The calibrated multinomial logistic regression implementation of HIMS-CDI (HIMS-CDI_cal) was quantised to INT8 precision using Post-Training Quantisation techniques to reduce computational overhead and memory consumption.

The remaining models (GNN, Transformer, and Edge Fusion variants) were not physically deployed on the Raspberry Pi platform. Their latency and memory statistics are reproduced from the primary evaluation workstation solely for comparative completeness.

Methodology (HIMS-CDI)

The HIMS-CDI_cal model was evaluated using the 48-sample synthetic test set comprising:

- 16 DEPLOY_SYNTH samples
- 32 NOT_DEPLOY_SYNTH samples

as described previously in Section 4.1 of the main manuscript.

To ensure timing stability and minimise runtime variance:

- A warm-up phase consisting of 50 inference passes was first executed.

- The complete test set was then inferred 100 consecutive times.
- Peak memory consumption was measured using the memory_profiler package.
- No additional user-space applications were active during benchmarking.

Results

Model	Inference Time (ms)	Peak RAM (MB)	Latency $\leq$ 200 ms	Memory $\leq$ 100 MB	Notes
HIMS-CDI_cal	132 $\pm$ 8 (single-thread) 164 $\pm$ 11 (5 concurrent)	91–96	✓	✓	Measured directly on Raspberry Pi 4
GNN	310 $\pm$ 17	420 $\pm$ 24	✗	✗	Workstation estimate reproduced from Table 4
Transformer	420 $\pm$ 22	512 $\pm$ 34	✗	✗	Workstation estimate
Edge-Fusion (original)	390 $\pm$ 20	405 $\pm$ 22	✗	✗	Workstation estimate
Edge-Fusion (optimised)	215 $\pm$ 15	98 $\pm$ 9	✗	✓ (borderline)	Workstation estimate

Interpretation

All Raspberry Pi deployment measurements obtained for HIMS-CDI_cal fall within the provisional governance gate thresholds defined by the taxonomy:

- **Latency:** < 200 ms
- **Memory Consumption:** < 100 MB

These findings confirm that the framework’s designated edge-ready model is practically deployable on representative clinical edge hardware. Consequently, the empirical deployment results strongly support the validity of the taxonomy’s “Edge Readiness” dimension and reinforce the suitability of lightweight calibrated linear architectures for constrained healthcare environments.

Conclusion

The edge benchmarking analysis demonstrates that HIMS-CDI_cal satisfies clinically relevant deployment constraints on a low-power edge platform while maintaining operational feasibility under concurrent inference conditions. In contrast, the more computationally intensive deep learning models substantially exceed the proposed governance thresholds. These findings therefore strengthen the manuscript’s broader conclusion that lightweight, interpretable architectures remain preferable for highly resource-constrained clinical edge deployments.